

Investigating with commands

NUnix · L3 MIAGE · CM2 · 90 minutes

ISTIC · University of Rennes 2026–2027



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Investigating with commands



A computer is slow. Would adding RAM help?
Check which resource is under pressure first.

1. Check CPU, memory and disk usage.
2. Find large files and inspect application logs.
3. Count error codes with a pipeline.

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1. Processes and resources



Check CPU, memory and disk usage

CPU, memory and disk space



Resource	Question	First observation
CPU	Is computation keeping a processor busy?	top, ps
RAM	Is memory pressure affecting the system?	free -h
Disk	Can the filesystem store more data?	df -h .

Adding RAM does not create free space on a full disk.

Running in the foreground



A **process** is a running instance of a program.

```
sleep 10
```

The shell waits for this foreground command to finish before showing a new prompt.

Ctrl+C requests an interrupt of the foreground job.

Inspecting processes



```
ps
ps -ef
```

Field	Meaning
PID	Process identifier
TTY	Associated terminal, if any
TIME	Accumulated CPU time
CMD	Command

Checking memory usage



```
free -h
ps -eo pid,comm,%cpu,%mem --sort=-%mem
```

- ▶ Read **available** memory; cache use is normal.
- ▶ Identify the application’s process and follow it over time.
- ▶ Check logs before attributing every delay to memory.

Checking disk usage



Tool	Scope	Question
df	A filesystem	How much space is available?
du	A directory tree	Which directories use space?
find	Individual entries	Which files fit my criteria?

Start broad, then inspect a **specific file**.

Find which directories use space



```
df -h "$HOME"
du -h --max-depth=1 "$HOME"
```

df shows usage of the filesystem containing your home.
du shows how much space your directories use.

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Finding files



```
find . -type f -name '*.txt'
```

- ▶ . is the **starting directory**.
- ▶ find also visits its subdirectories.
- ▶ -type f selects regular files.
- ▶ -name '*.txt' tests each entry's name.

The quotes let **find** receive the pattern unchanged.

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Search criteria



Criterion	Matches
-iname '*.txt'	Name, ignoring case
-user alice	Entries owned by Alice
-type d / -type l	Directories / symbolic links
-size +100M	Size greater than 100 MiB
-mtime -2	Modified less than 48 hours ago

```
find . -type f -size +100M
```

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Checkpoint: find



1. Find regular .c files below the current directory.
 2. Why quote '*.c'?
1. `find . -type f -name '*.c'`
 2. So the shell passes the pattern to find without expanding it first.

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2. Searching text



Read log records and search for errors

A small log file



Save these four lines as app.log:

```
10:00 INFO STARTED
10:01 ERROR DISK_FULL
10:02 ERROR TIMEOUT
10:03 ERROR DISK_FULL
```

Field	Meaning
1	Time
2	Severity
3	Event code

Finding text with grep



```
grep 'ERROR' app.log
grep -r 'ERROR' .
grep -n 'DISK_FULL' app.log
```

- ▶ Print **matching lines** from a file.
- ▶ -r searches directories recursively.
- ▶ -n includes line numbers.

Exit status: **0** = match, **1** = no match, **2** = error in GNU grep.

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Where the name grep comes from



The editor ed had the command pattern:

```
g/re/p
```

- ▶ **g**: apply globally to matching lines.
- ▶ **re**: a regular expression.
- ▶ **p**: print those lines.

grep makes that text search available as a separate command.

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Patterns in grep



Regular expression	Meaning
.	Any one character
m*	Zero or more repetitions of m
[1-3]	A digit from 1 to 3

```
grep 'm.*t' names.txt
grep 'toto[1-3]' names.txt
```

In a regular expression, * repeats the **preceding item**.

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Who interprets this pattern?



```
printf '%s\n' *.log
grep 'DISK.*' app.log
find . -name '*.log'
```

1. The **shell** expands a filename pattern.
2. **grep** interprets a regular expression.
3. **find** interprets a quoted filename pattern.

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Input, output and status



Channel	Number	Usual terminal behaviour
stdin	0	Read keyboard input
stdout	1	Display normal output
stderr	2	Display diagnostic messages

The **exit status** is a number reported when a command finishes. It is separate from the text it prints.

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Exit status



```
true
echo "$?"
false
echo "$?"
```

0 means success / true. **Nonzero** means failure / false.
\$? is the status of the **most recent command**.

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3. Redirections and pipes



Choose where a command reads and writes

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Reading input from a file



```
sort < names.txt
```

The shell opens `names.txt` as **standard input** for `sort`.

`sort` writes the sorted lines to standard output, which still points to the terminal.

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Saving output



```
sort names.txt > sorted.txt  
echo 'Zoé' >> sorted.txt
```

Operator	Effect on the destination file
>	Create it, or replace its contents
>>	Create it, or append to its contents

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Redirecting errors



```
find . -name '*.txt' > matches.txt 2> errors.txt
```

- ▶ > redirects **stdout**, descriptor 1.
- ▶ 2> redirects **stderr**, descriptor 2.
- ▶ Normal results and diagnostics go to separate files.

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Redirections are applied from left to right



```
ls notes.txt missing > all.txt 2>&1
ls notes.txt missing 2>&1 > out.txt
```

Command	stdout ends up...	stderr ends up...
First	In all.txt	In all.txt
Second	In out.txt	At the terminal

2>&1 copies stdout's destination **at that moment**.
It does not make stderr follow later changes to stdout.

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Why did the file become empty?



```
sort names.txt > names.txt
```

The shell opens and truncates names.txt **before** starting sort. The input is already gone.

Write to a **different file**, then inspect the result.

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A pipe between two commands



```
grep 'ERROR' app.log | wc -l
```

grep writes matching lines to its standard output.
The pipe connects that output to wc's standard input.

The result is the **number of matching lines**.

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Connecting programs like garden hoses



Doug McIlroy described connecting programs so that one program's output could become another's input.

```
grep 'ERROR' app.log | wc -l
```

Each tool has a small job.
The shared text stream lets us combine them.

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1. Select errors and extract their codes



```
grep ' ERROR ' app.log |  
  cut -d ' ' -f 3
```

```
DISK_FULL  
TIMEOUT  
DISK_FULL
```

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2. Count each error code



```
grep ' ERROR ' app.log |  
  cut -d ' ' -f 3 |  
  sort | uniq -c | sort -nr
```

```
2 DISK_FULL  
1 TIMEOUT
```

uniq -c counts **adjacent** equal lines. Sort first.

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Save and display output with tee



```
sort names.txt | tee sorted.txt
```

tee copies its input to a file **and** standard output.

Think of a T-shaped pipe fitting: the stream has two outlets.

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Combining arguments and redirections



```
sort < names.txt > sorted.txt  
sort < names.txt | head -n 3 > first-three.txt
```

- ▶ < names.txt selects the input stream.
- ▶ -n 3 remains an **argument** to head.
- ▶ > first-three.txt selects the output file.

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Checkpoint: redirections



- | | |
|---|-----------------------------------|
| 1. Which operator appends to a file? | 1. >> |
| 2. What does a b connect? | 2. a's stdout to b's stdin. |
| 3. Does a > out.txt also redirect standard error? | 3. No. Use 2> to redirect stderr. |

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The exit status of a pipeline



```
false | sort
echo "$?"
set -o pipefail
false | sort
echo "$?"
```

Output: **0**, then **1**.

Bash normally reports the last pipeline command's status.

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CM2 summary: commands and pipelines



- ▶ Check **CPU**, **memory** and **disk** usage.
- ▶ Find files and filter lines with **grep**.
- ▶ Combine commands with pipes and redirections.

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Optional reference



More search expressions and interactive process control

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Combining criteria



```
find /usr -name '*.c'
find . -user alice -name '*.c'
find . -not \( -name '*.obj' -o -name '*.map' \)
```

- ▶ Adjacent tests use **AND**. -a makes it explicit.
- ▶ -o means **OR**. -not negates a test.
- ▶ Escaped parentheses group tests for find.

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Running a command on matches



```
find . -type f -name '*.txt' -exec wc -l {} \;
```

- ▶ -exec starts the command to run.
- ▶ {} stands for the current matching path.
- ▶ \; ends the command without ending the shell's command line.

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Running in the background



```
sleep 60 &  
jobs  
fg %1
```

- ▶ & lets the shell return to the prompt immediately.
- ▶ jobs lists jobs managed by this shell.
- ▶ fg %1 brings job **1** to the foreground.

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Suspending and resuming a job



Action	Effect
Ctrl+Z	Suspend the foreground job
bg %1	Resume job 1 in the background
fg %1	Resume job 1 in the foreground

A **job number** belongs to a shell.
A **PID** identifies a process in the system.

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Sequential and asynchronous execution

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```
who | sort; date
sleep 10 &
date
```

- ▶ ; runs the next command after the previous one finishes.
- ▶ & starts work asynchronously.
- ▶ The final date can run while sleep is still active.

Signals and end of input

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Action	Meaning
kill PID	Send SIGTERM to a process
Ctrl+C	Send SIGINT to the foreground process group
Ctrl+\	Send SIGQUIT to the foreground process group
Ctrl+Z	Send SIGTSTP to suspend the foreground group
Ctrl+D	Terminal end-of-input action